SYSTEMS ANALYSIS & REQUIREMENTS ENGINEERING

Medical Lab Test Ordering and Delivery System

Deliverable: 20 Questions (10 Interview + 10 Web Form) & System Analysis Report

Methodology: Mixed-Methods (Qualitative & Quantitative)

SECTION 1: OPEN-ENDED QUESTIONS (10 STAKEHOLDER INTERVIEW QUESTIONS)

Target Audience: Laboratory Managers, Diagnostic Technicians, Customer Support Receptionists, and Field Phlebotomists. Designed to gather qualitative workflow insights.

Q1 Can you walk me through the current step-by-step workflow from the moment a patient requests a test until they receive their results?

> **Answer:** Currently, a patient either walks into the branch or calls us via phone or WhatsApp to book a home visit. The receptionist writes down their details manually or logs them into a basic local Excel sheet, checks the test prices, and schedules a phlebotomist. After the sample is drawn and analyzed in the lab, the result is printed as a hard copy or saved as a PDF and sent manually via WhatsApp.

Q2 What are the biggest operational bottlenecks or challenges you face with home sample collection requests?

> **Answer:** The biggest issues are inaccurate home addresses communicated over the phone, poor route coordination among our phlebotomists—which delays visits and risks sample integrity—and communication gaps regarding test pre-conditions, such as a phlebotomist arriving at a patient's home only to discover the patient did not fast for the required 8 to 12 hours.

Q3 How do you currently handle medical prescriptions uploaded or brought in by patients whose physician handwriting is unclear?

> **Answer:** Patients send a photograph of the handwritten prescription via WhatsApp. A senior lab receptionist or clinical doctor reviews it to decipher the required diagnostic tests, calculates the total cost, and replies to the patient. However, incoming messages frequently get buried under dozens of active chats, causing slow response times and lost bookings.

Q4 How are collected blood and biological samples tracked from the patient's home until they reach the laboratory analyzers?

> **Answer:** Right now, the field phlebotomist writes the patient's full name and phone number on the collection tube label by hand. This manual labeling occasionally leads to human error, smudge marks, or clerical mix-ups before the sample receives an official printed barcode inside the main laboratory branch.

Q5 What safety and timing constraints must the system enforce for transporting biological samples from homes to the laboratory?

> **Answer:** Certain diagnostic tests—such as coagulation profiles, arterial blood gases, or specific hormone assays—are highly time- and temperature-sensitive. They must reach the lab within 1 to 2 hours of collection inside calibrated cooling boxes. The system requires a priority timer to alert couriers and lab staff about urgent samples.

SECTION 1 (CONTINUED): OPEN-ENDED STAKEHOLDER INTERVIEW QUESTIONS

Q6 **How do you currently notify patients or physicians when critical or abnormal test results are detected?**

> **Answer:** If an assay reveals a life-threatening or critical panic value, the laboratory specialist calls the patient or their referring physician directly by telephone to verify clinical history before releasing the standard digital report.

Q7 **What difficulties do your staff face when managing test catalogs, pricing updates, and medical insurance discounts?**

> **Answer:** Updating test prices or insurance coverage rules across multiple branches is chaotic because receptionists rely on printed price lists or unlinked spreadsheets. Whenever an insurance provider changes its co-pay percentage or pre-approval policy, billing errors happen frequently during checkout.

Q8 **How do patients currently receive their test results, and what complaints do they raise about this delivery process?**

> **Answer:** Most patients must either visit the branch physically to pick up printed results or message us repeatedly on WhatsApp asking if their reports are ready. They frequently complain about waiting on hold, delayed WhatsApp replies, and losing old paper records when they need to compare historical health trends.

Q9 **What financial and administrative reports do you need the new system to generate regularly?**

> **Answer:** We require automated daily revenue reports segmented by payment method (cash, credit card, mobile wallets, insurance), a breakdown of the most requested lab tests, phlebotomist performance metrics (successful home visits per shift), and aged receivables from corporate or insurance partners.

Q10 **What are your main expectations and security requirements if we transition this entire workflow to an online platform?**

> **Answer:** First, strict medical data privacy—patient records and test results must be encrypted and accessible only by the authenticated patient and authorized lab personnel. Second, an automated booking flow that displays test prerequisites (such as fasting hours) clearly upon checkout, combined with live order status tracking so patients stop calling reception to check result readiness.

10	10	1–2 hrs	100%
Staff Interviews (Open-Ended)	Web Form Items (Closed-Ended)	Max Sample Transit Window	Barcode & Digital Traceability

SECTION 2: CLOSED-ENDED QUESTIONS (10 WEB FORM / QUESTIONNAIRE QUESTIONS)

Target Audience: Patients and End-Users. Designed as structured multiple-choice questions for Google Forms to collect quantitative market statistics.

Live Google Form URL: *https://docs.google.com/forms/d/e/1FAIpQLSc5OQ0QlbuiL97PewkYAt5xF9WuELsJjy2YAq9atYHiGqn1kg/viewform?usp=publish-editor*

Q11 **Have you ever ordered a medical lab test via a website or mobile application before?**

[ ] Yes (32%) **[✓] No (68% — Selected Majority)**

Q12 **How do you usually prefer to have your medical samples collected?**

[ ] Visit the lab branch (25%) **[✓] Home visit sample collection (55% — Majority)** [ ] Depends on health condition (20%)

Q13 **Do you often need to book lab tests by uploading a doctor's handwritten prescription image rather than selecting test names manually?**

[✓] Yes, always (62% — Majority) [ ] Sometimes (26%) [ ] No, I know the test names (12%)

Q14 **Which method do you prefer for receiving your final medical lab results?**

[✓] In-app / Website PDF Download (60% — Majority) [ ] WhatsApp / Email (30%) [ ] Printed Hard Copy to Home (6%) [ ] Branch Pick-up (4%)

Q15 **What is the maximum acceptable waiting window for a technician to arrive at your home for sample collection?**

[ ] Within 1 hour (12%) [ ] 1–2 hours (10%) **[✓] Specific scheduled time slot / 30-min window (74% — Majority)** [ ] Same day anytime (4%)

Q16 **Would you find it helpful to receive automated notifications regarding pre-test preparation instructions (e.g., fasting hours, water intake)?**

[✓] Very helpful (89% — Majority) [ ] Somewhat helpful (9%) [ ] Not needed (2%)

Q17 **How would you prefer to pay for your lab test orders and home visits?**

[✓] Mobile Wallets / InstaPay (40% — Tied Majority) **[✓] Cash on collection (40% — Tied Majority)** [ ] Medical Insurance Card (12%) [ ] Credit/Debit Card (8%)

Q18 **Would you like the system to store a historical archive of all your past lab results with visual trend charts (e.g., tracking blood sugar over months)?**

[✓] Yes (92% — Majority) [ ] No (8%)

Q19 **How important is real-time tracking (similar to delivery apps) for knowing when the phlebotomist is on their way and what stage your sample is in?**

[✓] Extremely important (81% — Majority) [ ] Moderately important (15%) [ ] Not important (4%)

Q20 **Are you willing to pay an additional service fee for express home sample collection and home delivery of printed reports?**

[✓] Yes, if the fee is reasonable (64% — Majority) [ ] Only in emergency cases (25%) [ ] No, only if free (11%)

Questionnaire Dimension	Questions	Primary Quantitative Finding	Direct System Feature Mapped
Ordering & Rx Upload	Q11, Q13, Q16	62% upload handwritten Rx; 89% want prep alerts	Prescription Image OCR/Review & Fasting Push Alerts

SECTION 3: COMPREHENSIVE SYSTEM ANALYSIS REPORT

1. INTERVIEW OPINION (ANALYST ASSESSMENT)

Based on the interviews conducted with laboratory management, receptionists, and field phlebotomists, it is evident that the laboratory possesses strong diagnostic expertise but suffers severely from **fragmented communication and manual administrative workflows**. From a systems analyst perspective, relying on phone calls and WhatsApp messages to handle complex medical bookings, prescription decoding, and courier dispatching is unsustainable and unscalable. The absence of a centralized platform connecting the patient, the field phlebotomist, and the internal Laboratory Information System (LIS) creates a high risk of human error—specifically regarding handwritten sample labels, missed fasting instructions, and delayed result delivery. Implementing an automated ordering and delivery system is an operational necessity to eliminate bottlenecks and reduce staff burnout.

2. INTERVIEW FEELINGS (STAKEHOLDER ATTITUDES & EMOTIONAL RESPONSES)

- **Frustration & Exhaustion (Reception & Support Staff):** Receptionists expressed overwhelming stress from answering hundreds of repetitive phone calls and WhatsApp chats daily regarding test prices, fasting rules, and result readiness.
- **Anxiety Over Operational Risk (Field Phlebotomists):** Home-visit technicians felt frustrated by inaccurate phone addresses and wasted trips when patients failed to fast. They also felt pressured by strict transit time limits for sensitive samples without route optimization.
- **Enthusiasm & Optimism (Laboratory Management):** Lab directors were highly cooperative and eager for digital transformation, viewing the proposed platform as a major competitive advantage to expand home-visit market share.
- **Mild Apprehension (Senior Technicians):** A few traditional staff members showed slight hesitation about learning new software, emphasizing that the interface must be intuitive and not slow down daily bench work.

3. CURRENT STATE OF THE SYSTEM (AS-IS SYSTEM)

- **Ordering & Inquiry:** Patients visit a branch, call a hotline, or send WhatsApp photos of handwritten prescriptions that often get buried in chat queues.
- **Scheduling & Dispatching:** Home visits are logged manually into local Excel sheets or paper ledgers; phlebotomists are dispatched via phone without GPS pinning or route planning.
- **Sample Collection & Labeling:** Phlebotomists write patient identifiers by hand on collection tubes at the patient's home and collect cash using paper receipts.
- **Processing & Result Delivery:** Patient data is re-entered manually at the lab. Completed reports are printed for branch pick-up or exported manually as PDFs and sent over WhatsApp without a unified patient archive.

4. SYSTEM GOALS (TO-BE SYSTEM)

- **Automate Booking & Prescription Upload:** Provide a web/mobile portal for browsing test catalogs, transparent pricing, insurance validation, and rapid prescription image verification.
- **Enforce Pre-Test Compliance:** Automatically display mandatory preparation rules (fasting hours, medication holds) at checkout and send automated reminder notifications prior to collection.
- **Optimize Home Logistics & Barcode Traceability:** Integrate GPS pinning, 30-minute time-slot booking, portable barcode generation at the point of collection, and cold-chain timers for urgent samples.
- **Real-Time Order Tracking & Digital Result Delivery:** Enable live stage tracking (*Confirmed → Phlebotomist En Route → Sample Collected → In Lab → Results Ready*) and instant encrypted PDF downloads with historical health trend charts, plus optional hard-copy courier delivery.